

Tasks:

- 1. Open Burp Suite.**
- 2. Navigate to Decoder.**
- 3. Perform the following operations:**
 - **URL Encode**
 - **URL Decode**
 - **Base64 Encode**
 - **Base64 Decode**
 - **HTML Encode**
 - **HTML Decode**
- 4. Take screenshots of each conversion.**
- 5. Create 5 sample strings and perform all encoding and decoding operations.**

Examples:

- **CyberSecurity**
 - **VExTechSolution**
 - **Password123**
 - **Student01**
 - **HelloWorld**
- 6. Navigate to Comparer.**
 - 7. Compare:**
 - **Two different login requests**
 - **Two different responses**
 - **Two different encoded values**
 - 8. Observe and note:**
 - **Similarities**
 - **Differences**
 - **Modified parameters**

Assignment:

Prepare a report containing:

- 9. What is Encoding?**
- 10. Difference between Encoding, Encryption and Hashing.**
- 11. What is Base64?**
- 12. What is URL Encoding?**
- 13. Why do web applications use encoding?**
- 14. Screenshots of Decoder operations.**
- 15. Screenshots of Comparer results.**

Using Burp Decoder, decode the following values and explain the output:

- **SGVsbG8gV29ybGQ=**
- **Q3liZXIgaU2VjdXJpdHk=**
- **VkVYIFRIY2ggU29sdXRpb24=**
- **UGFzc3dvcmQxMjM=**

Introduction

Burp Suite is a popular web application security testing tool used by penetration testers and cyber security professionals. It provides multiple tools to inspect, analyze, modify, encode, decode, and compare HTTP requests and responses.

In this practical, two Burp Suite modules are used:

1. Decoder
2. Comparer

What is Encoding?

Encoding is the process of converting data from one format into another format so that it can be transmitted or stored properly.

Examples:

- Base64 Encoding
- URL Encoding
- HTML Encoding

Difference Between Encoding, Encryption, and Hashing

Feature	Encoding	Encryption	Hashing
Purpose	Data conversion	Data protection	Data verification
Reversible	Yes	Yes (with key)	No
Security	No	Yes	One-way
Key Required	No	Yes	No
Examples	Base64	AES, RSA	SHA-256, MD5

What is Base64 Encoding?

Base64 is a method of converting data into readable text format using letters, numbers, and symbols.

It is widely used in:

- API communication
- Authentication
- Email attachments
- Web tokens

Example:

Original:Hello

Base64:SGVsbG8=

What is URL Encoding?

URL Encoding converts special characters into URL-safe format.

Example:

- Space → %20
- @ → %40
- / → %2F

Example:

Original:Hello World

Encoded:Hello%20World

Why Do Web Applications Use Encoding?

Web applications use encoding for:

- Safe data transmission
- Sending special characters
- Browser compatibility
- Avoiding data corruption
- Transferring binary data
- Displaying HTML properly

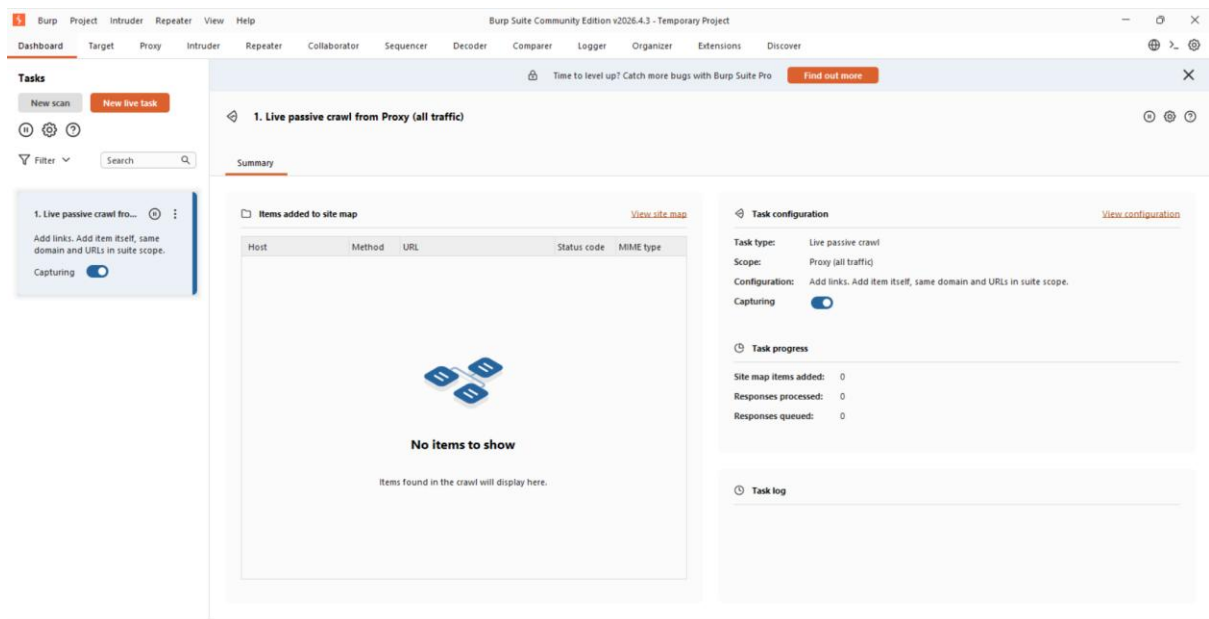
PART A — BURP SUITE DECODER

Step 1: Open Burp Suite

Procedure

1. Open Burp Suite Community Edition.
2. Click on:Start Burp
3. Wait for the dashboard to load.

Screenshot

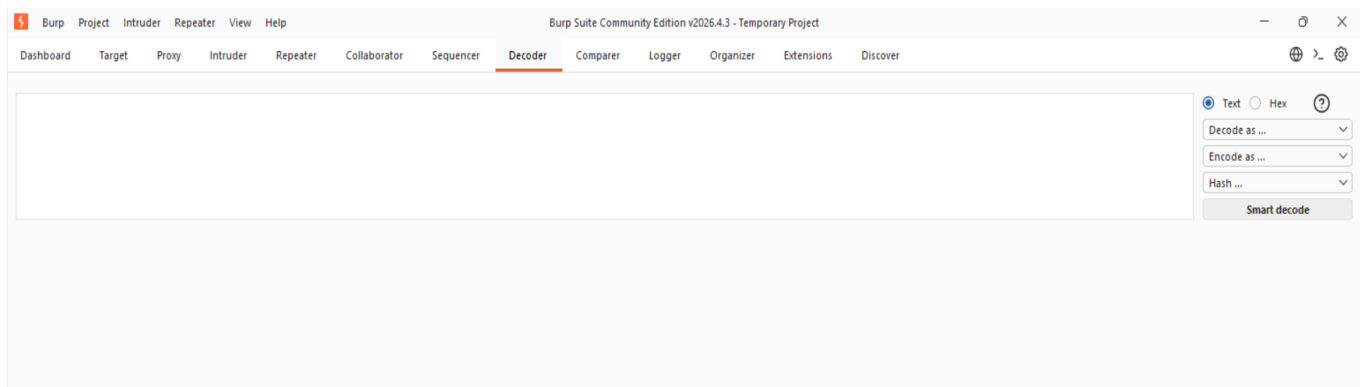


Step 2: Open Decoder Tab

Procedure

1. From the top menu bar click:Decoder
2. Decoder window will open.

Screenshot



String 1: CyberSecurity

Step 3: Enter Sample String

Procedure

1. In the Decoder input box type: CyberSecurity
2. The text will appear in the input section.

Step 4: Perform URL Encode

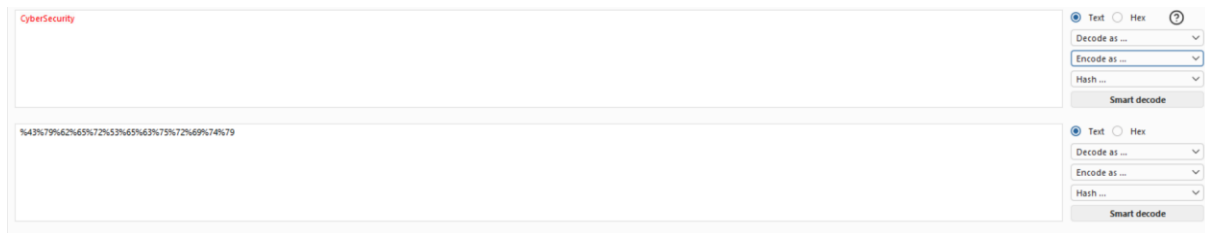
Procedure

1. Select the text.
2. Right click.
3. Click: Encode as URL

OR

Use the Encode dropdown menu.

Screenshot

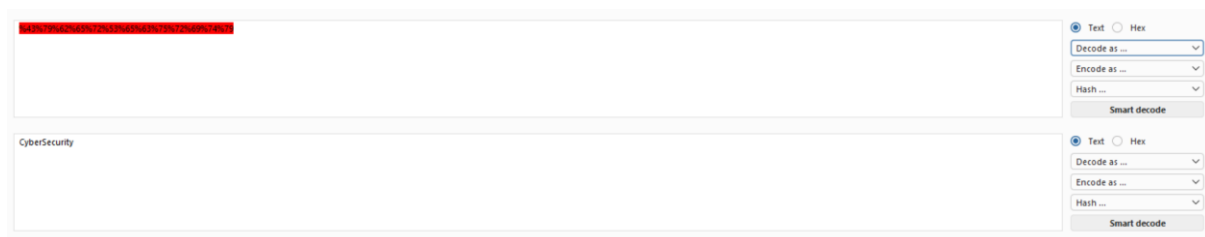


Step 5: Perform URL Decode

Procedure

1. Select encoded text.
2. Right click.
3. Click:
Decode as URL

Screenshot

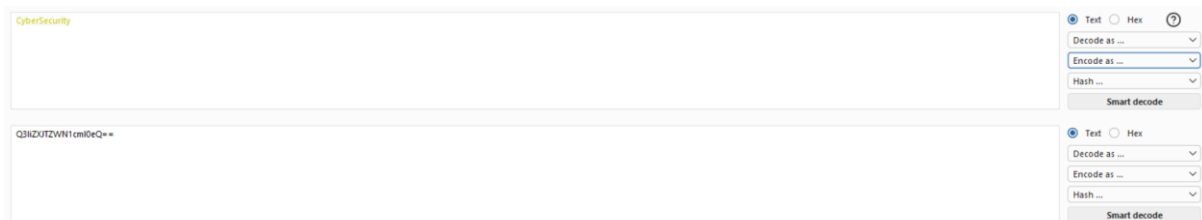


Step 6: Perform Base64 Encode

Procedure

1. Select the original text.
2. Right click.
3. Click:
Encode as Base64

Screenshot

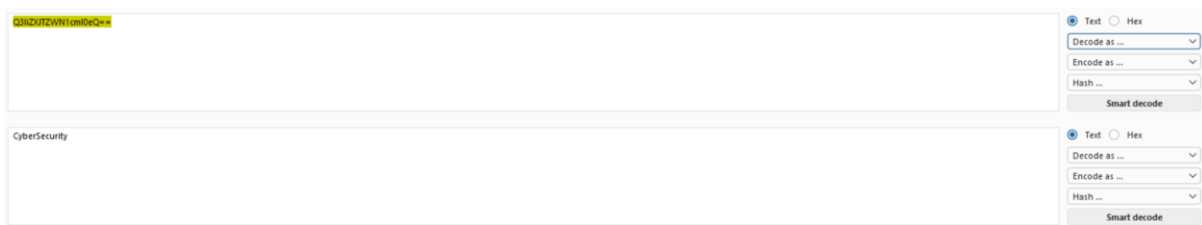


Step 7: Perform Base64 Decode

Procedure

1. Select Base64 encoded value.
2. Right click.
3. Click:
Decode as Base64

Screenshot

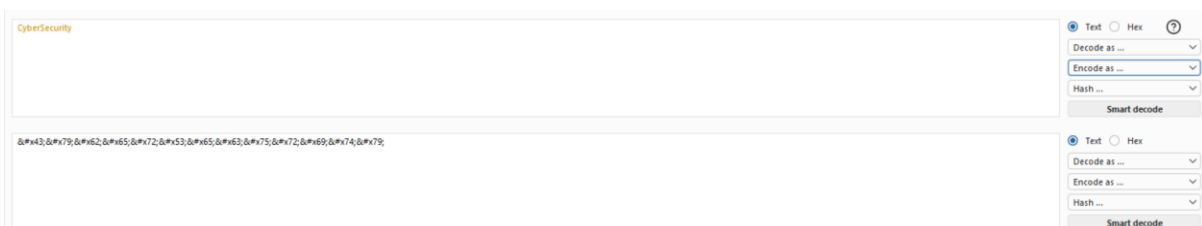


Step 8: Perform HTML Encode

Procedure

1. Select the text.
2. Right click.
3. Click:
Encode as HTML

Screenshot

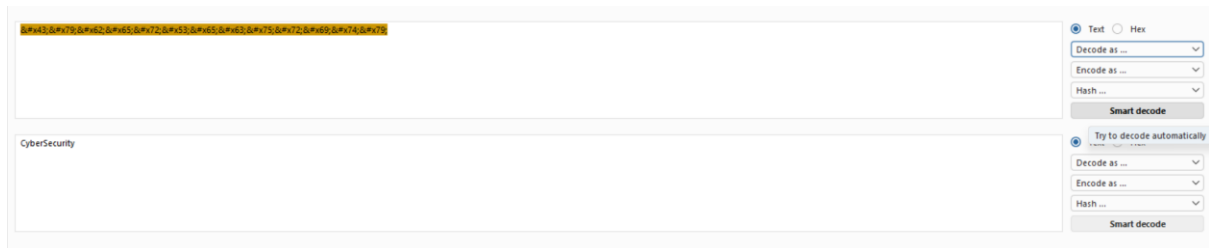


Step 9: Perform HTML Decode

Procedure

1. Select encoded HTML text.
2. Right click.
3. Click:
Decode as HTML

Screenshot



String 2: VExTechSolution

Step 1: Enter String in Decoder

Procedure

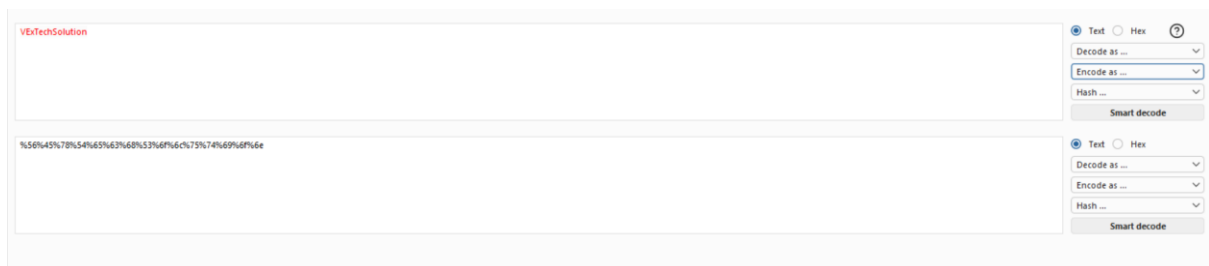
1. Open Burp Suite.
2. Navigate to the Decoder tab.
3. Enter the text:VExTechSolution

Step 2: URL Encode

Procedure

1. Select the text.
2. Right click.
3. Click:
Encode as URL

Screenshot

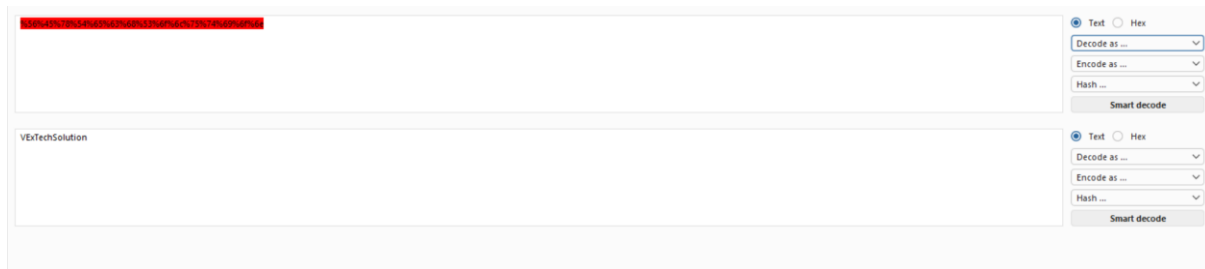


Step 3: URL Decode

Procedure

1. Select encoded value.
2. Right click.
3. Click:
Decode as URL

Screenshot

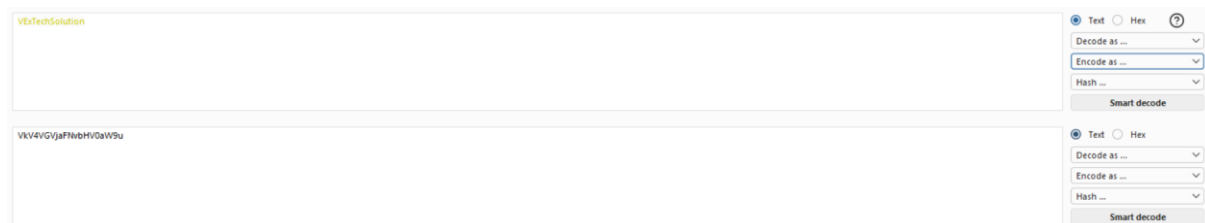


Step 4: Base64 Encode

Procedure

1. Select original text.
2. Right click.
3. Click:
Encode as Base64

Screenshot

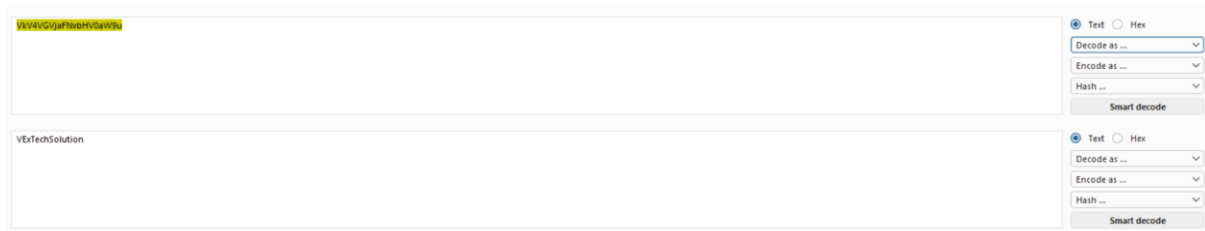


Step 5: Base64 Decode

Procedure

1. Select Base64 value.
2. Right click.
3. Click:
Decode as Base64

Screenshot

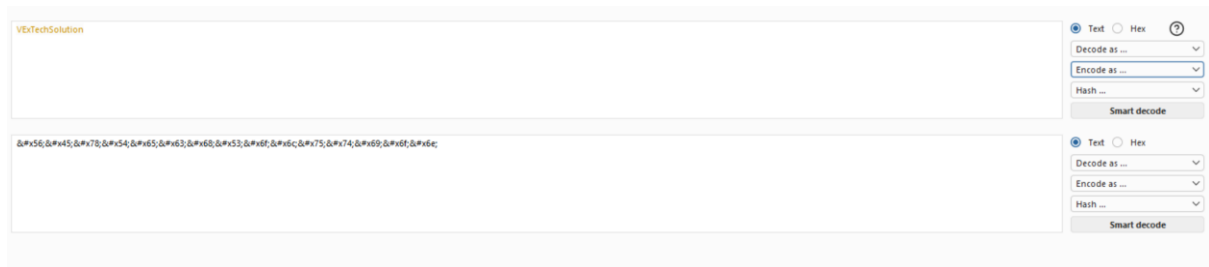


Step 6: HTML Encode

Procedure

1. Select text.
2. Right click.
3. Click:
Encode as HTML

Screenshot

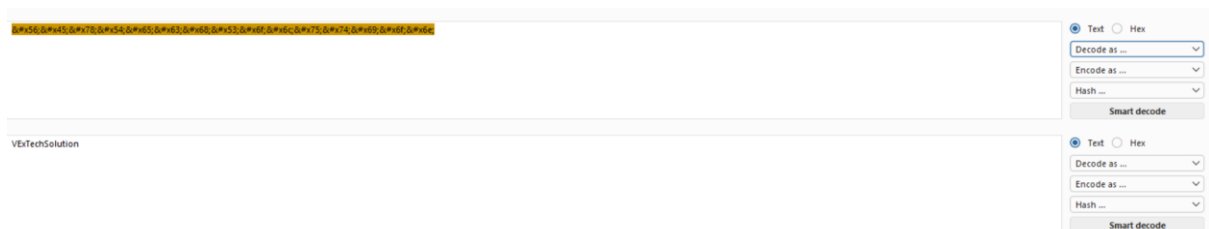


Step 7: HTML Decode

Procedure

1. Select HTML encoded text.
2. Right click.
3. Click:
Decode as HTML

Screenshot



String 3: Password123

Step 1: Enter String in Decoder

Procedure

1. Open Decoder tab.
2. Enter:
Password123

Step 2: URL Encode

Screenshot

Password123

%50%61%73%73%77%68%72%64%31%53%33

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Step 3: URL Decode

Screenshot

U50%61%73%73%77%68%72%64%31%53%33

Password123

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Step 4: Base64 Encode

Screenshot

Password123

UGFzc3dvcmQ6MTIz

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Step 5: Base64 Decode

Screenshot

UGFzc3dvcmQ6MTIz

Password123

Text

Hex

?

Decode as ...

Encode as ...

Hash ...

Smart decode

Text

Hex

?

Decode as ...

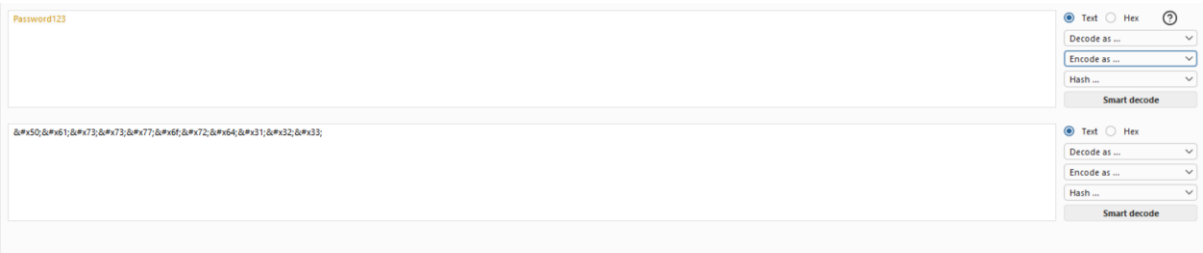
Encode as ...

Hash ...

Smart decode

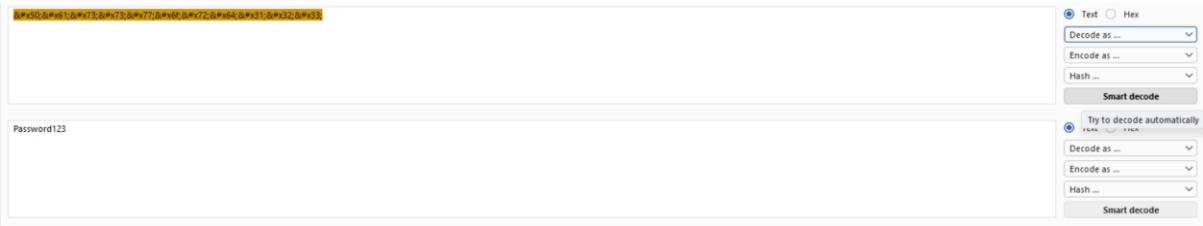
Step 6: HTML Encode

Screenshot



Step 7: HTML Decode

Screenshot



String 4: Student01

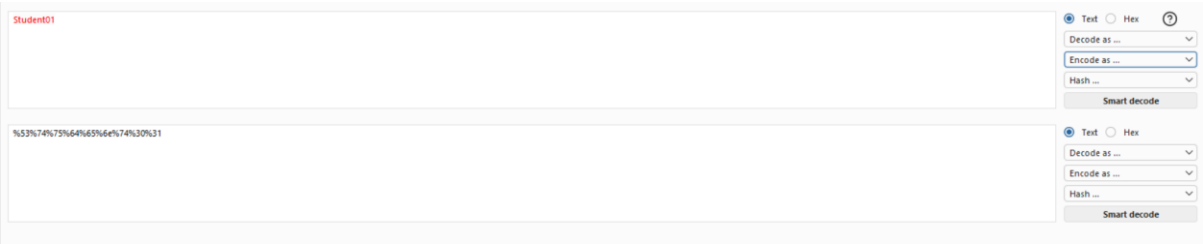
Step 1: Enter String in Decoder

Procedure

- 1. Open Decoder.
- 2. Enter:
Student01

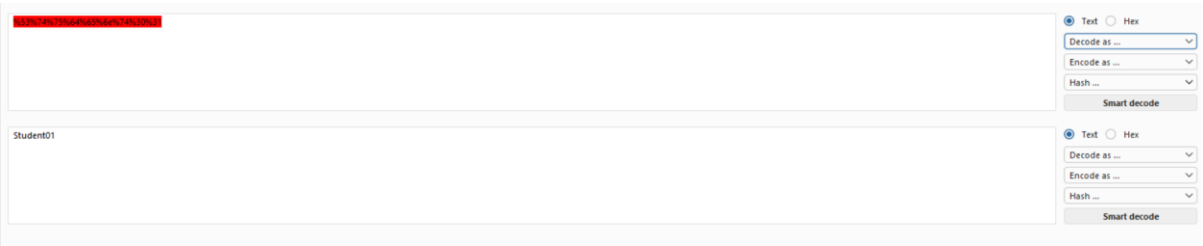
Step 2: URL Encode

Screenshot



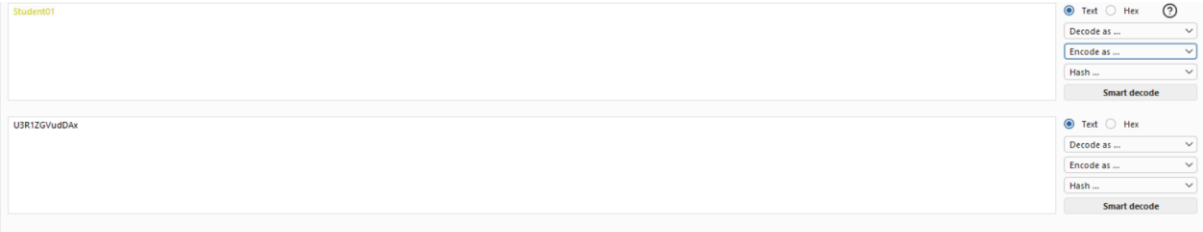
Step 3: URL Decode

Screenshot



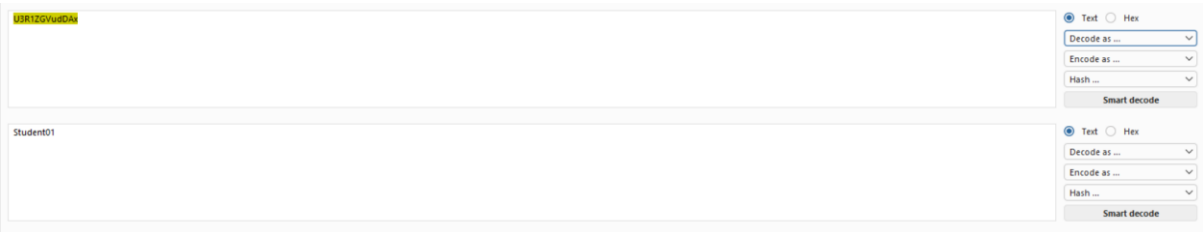
Step 4: Base64 Encode

Screenshot



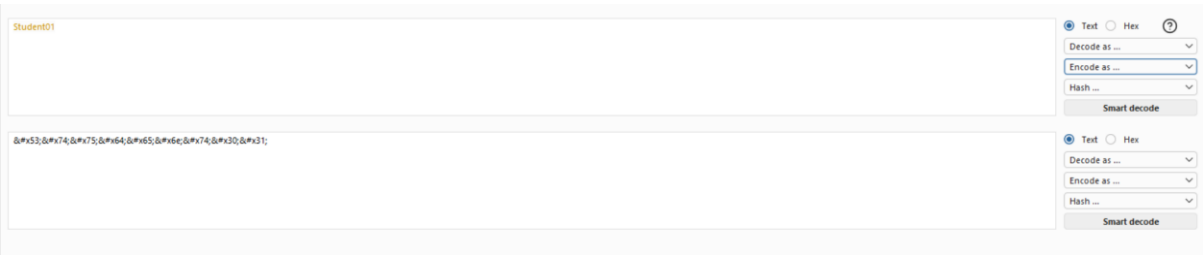
Step 5: Base64 Decode

Screenshot



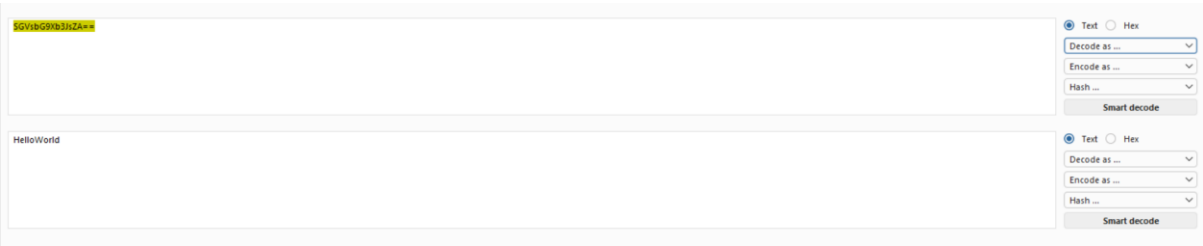
Step 6: HTML Encode

Screenshot



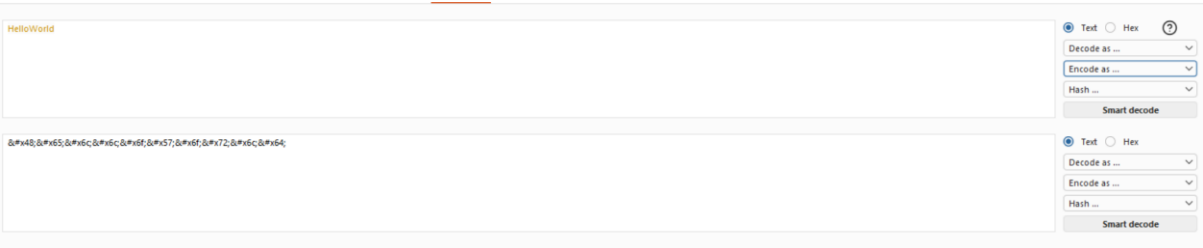
Step 5: Base64 Decode

Screenshot



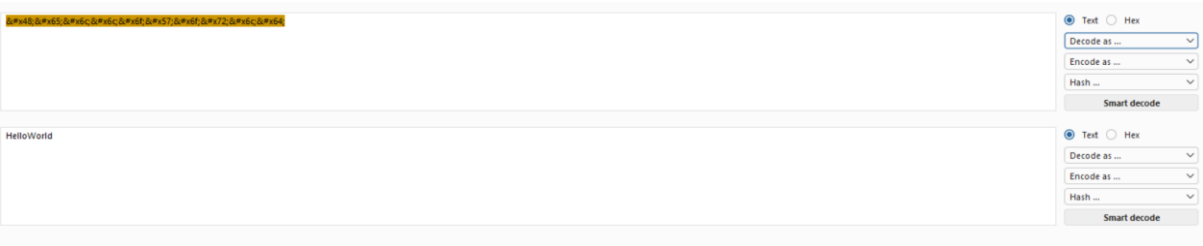
Step 6: HTML Encode

Screenshot



Step 7: HTML Decode

Screenshot



PART B — BURP SUITE COMPARER

Step 1: Capture Two Requests

Procedure

1. Open Burp Suite.
2. Go to: Proxy → Intercept
3. Turn:
Intercept ON
4. Open browser.
5. Visit any login page.
6. Enter different usernames and passwords twice.
7. Capture both requests.

Example:

- Request 1 → admin/admin123
- Request 2 → student/student123

Screenshot:

- Request 1 → admin/admin123

The screenshot displays the Burp Suite interface with a list of captured requests at the top. The selected request (index 19) is a POST to https://demo.testfire.net/doLogin. The main pane shows the 'Request' tab with a 'Pretty' view of the HTTP request. The request body contains the parameters uid=admin and pass=admin123. The 'Inspector' pane on the right shows the request attributes, including the protocol (HTTP/2), method (POST), and path (/doLogin). The request body parameters are also listed, showing uid=admin and pass=admin123.

- Request 2 → student/student123

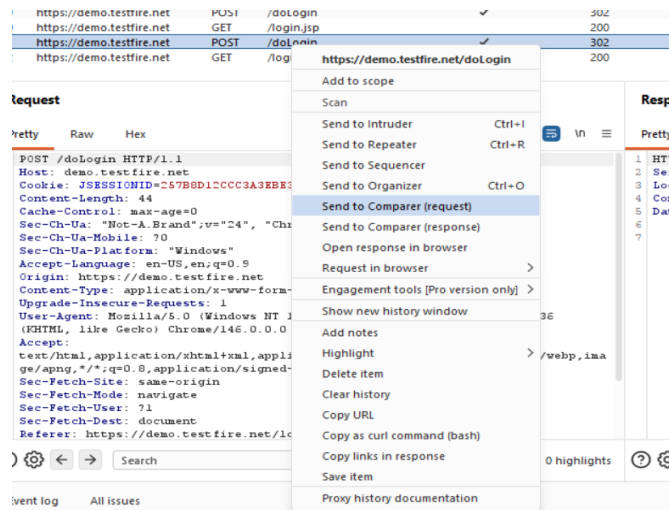
The screenshot displays the Burp Suite interface with a list of captured requests at the top. The selected request (index 22) is a POST to https://demo.testfire.net/doLogin. The main pane shows the 'Request' tab with a 'Pretty' view of the HTTP request. The request body contains the parameters uid=student and pass=student123. The 'Inspector' pane on the right shows the request attributes, including the protocol (HTTP/2), method (POST), and path (/doLogin). The request body parameters are also listed, showing uid=student and pass=student123.

Step 2: Send Requests to Comparer

Procedure

1. Right click first request.
2. Click:
Send to Comparer
3. Repeat for second request.

Screenshot

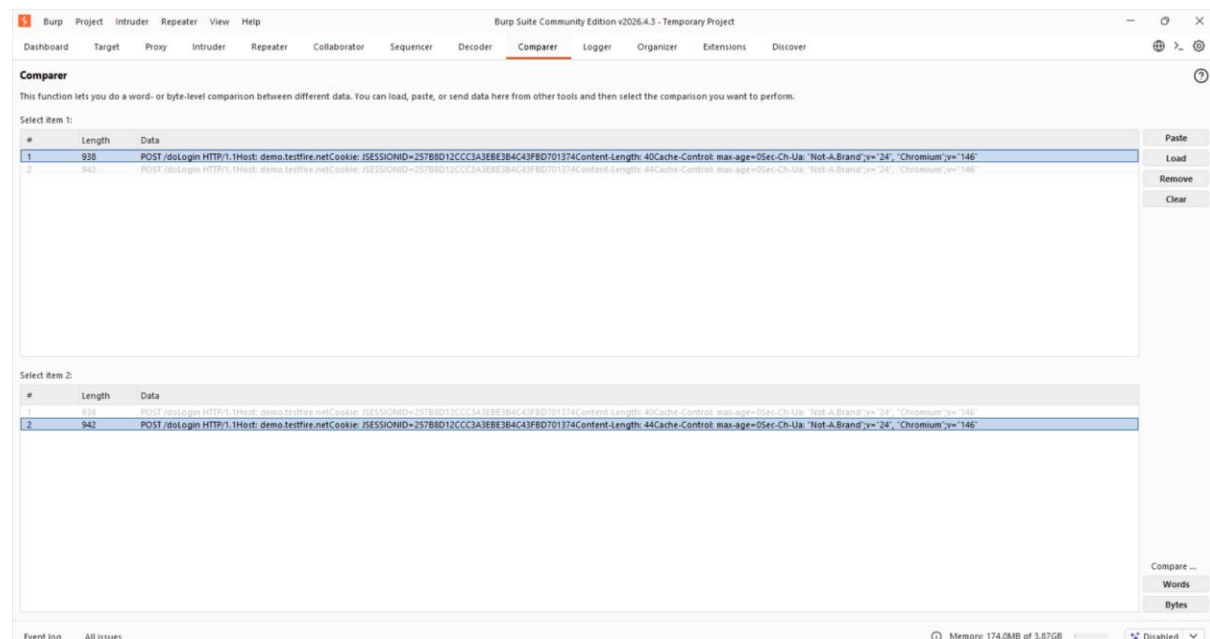


Step 3: Open Comparer Tab

Procedure

1. Click:
Comparer Tab
2. Both requests will appear.

Screenshot



Step 4: Compare Requests

Procedure

1. Select both requests.
2. Click:
Words Comparison

Observation

Similarities

- Same request method
- Same URL
- Same headers

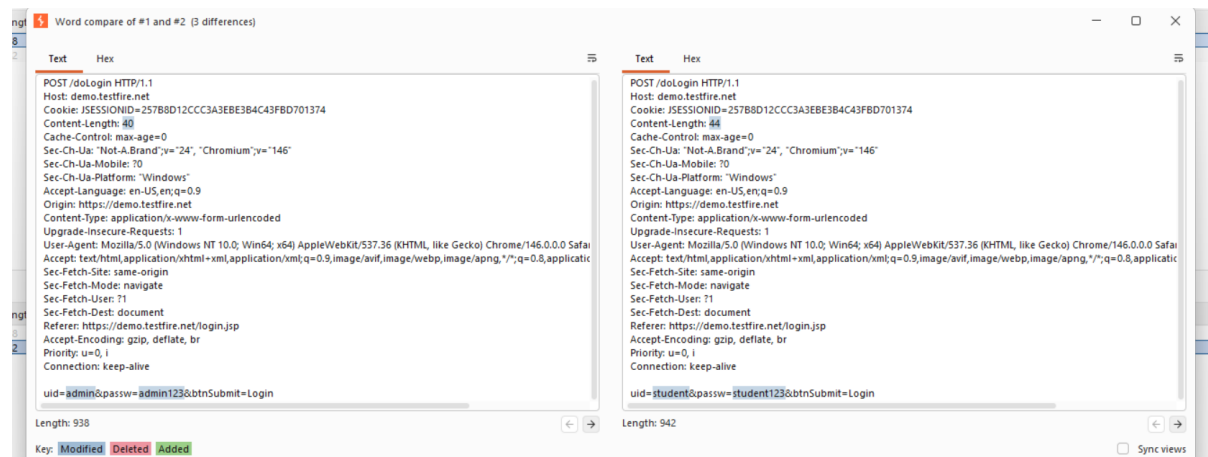
Differences

- Username changed
- Password changed
- Session cookie changed

Modified Parameters

- username
- password
- session value

Screenshot



Step 5: Compare Responses

Procedure

1. Capture two responses.
2. Send both to Comparer.
3. Compare using: Words Comparison

Observation

Similarities

- Both responses use the same protocol:
HTTP/1.1
- Both responses returned the same status code:
302 Found
- Same server information:
Apache-Coyote/1.1
- Same redirect location:
login.jsp
- Same content length

Differences

- Date and time values are different in both responses.
- First response time:
10:04:05 GMT
- Second response time:
10:06:53 GMT

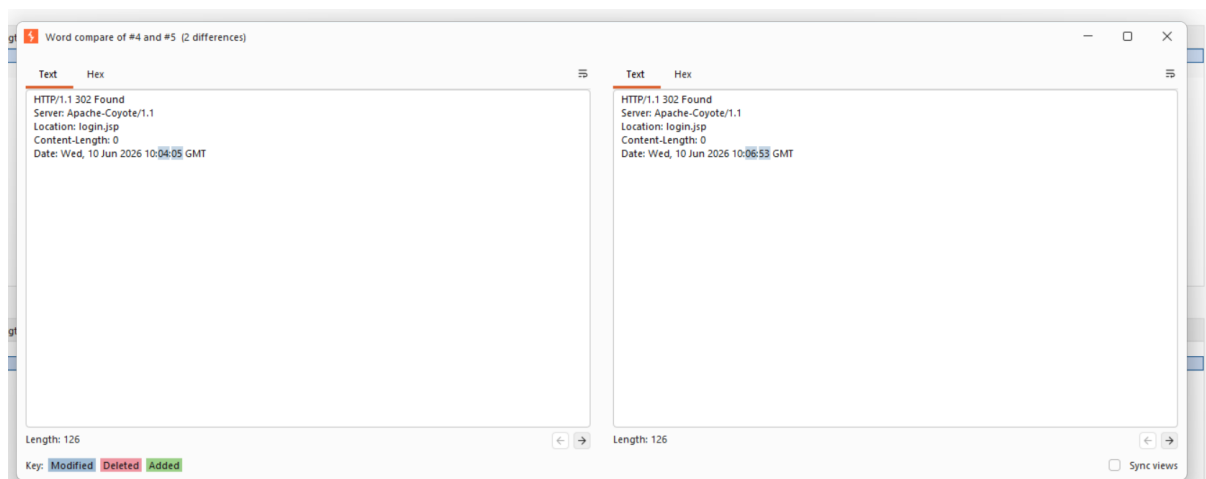
Modified Parameters

The modified parameter observed is:

- Date header value

No other headers or response parameters were modified

Screenshot



Step 6: Compare Two Encoded Values

Example Values

1. SGVsbG9Xb3JsZA==
2. UGFzc3dvcmQxMjM=

Observation

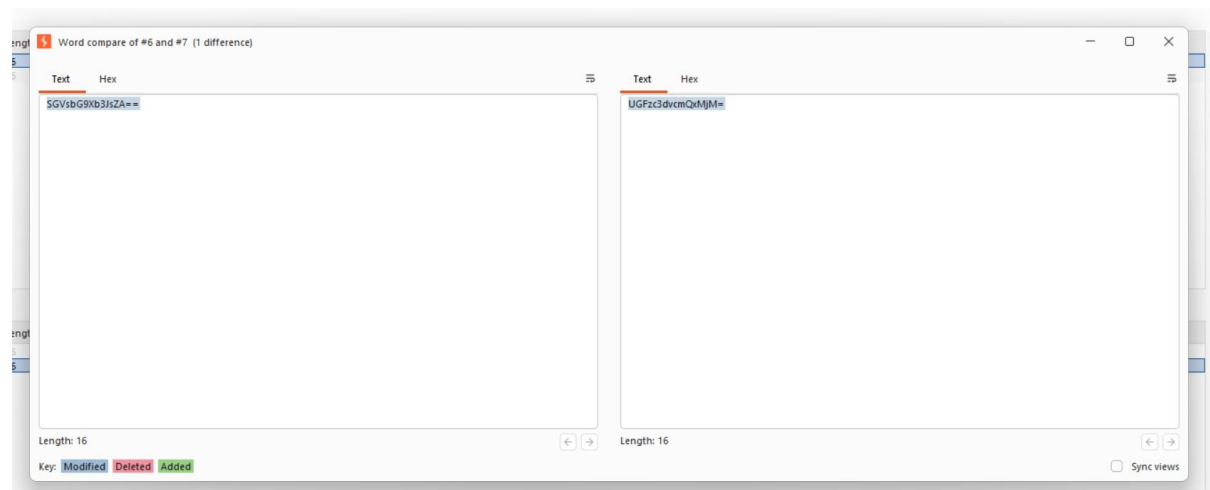
Similarities

- Both are Base64 encoded
- Both contain alphanumeric characters

Differences

- Encoded outputs differ

Screenshot



PART C — DECODING GIVEN VALUES

1. SGVsbG8gV29ybGQ=

Steps

1. Open Decoder.
2. Paste:
SGVsbG8gV29ybGQ=
3. Right click.
4. Click:
Decode as Base64

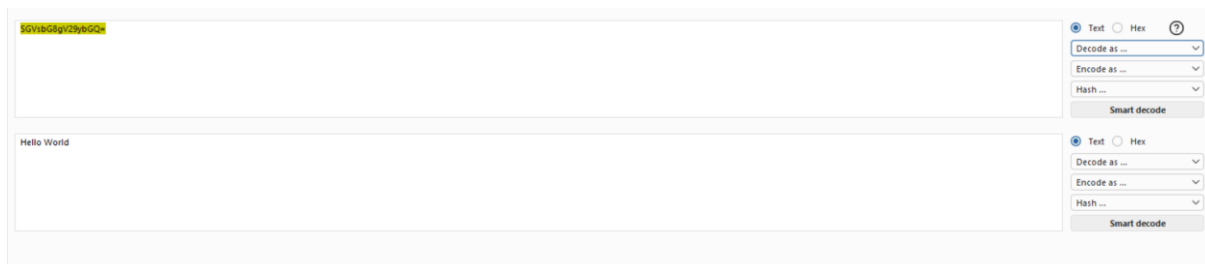
Output

Hello World

Explanation

The encoded Base64 value converts into readable text “Hello World”.

Screenshot



2. Q3liZXlIgU2VjdXJpdHk=

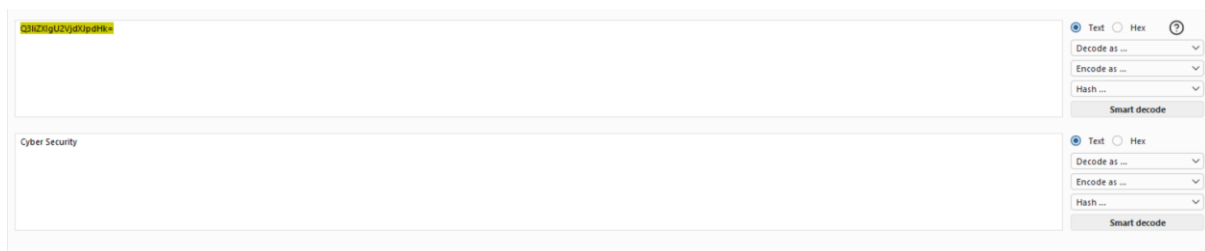
Output

Cyber Security

Explanation

The Base64 encoded text represents “Cyber Security”.

Screenshot



3. VkVYIFRIY2ggU29sdXRpb24=

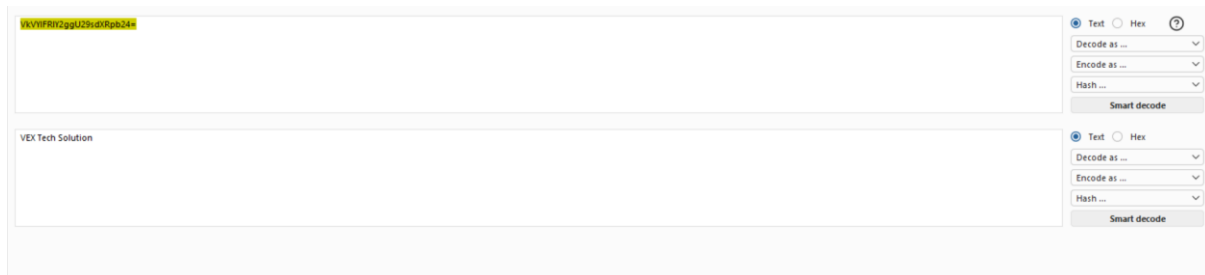
Output

VEX Tech Solution

Explanation

This encoded value decodes into “VEX Tech Solution”.

Screenshot



4. UGFzc3dvcmQxMjM=

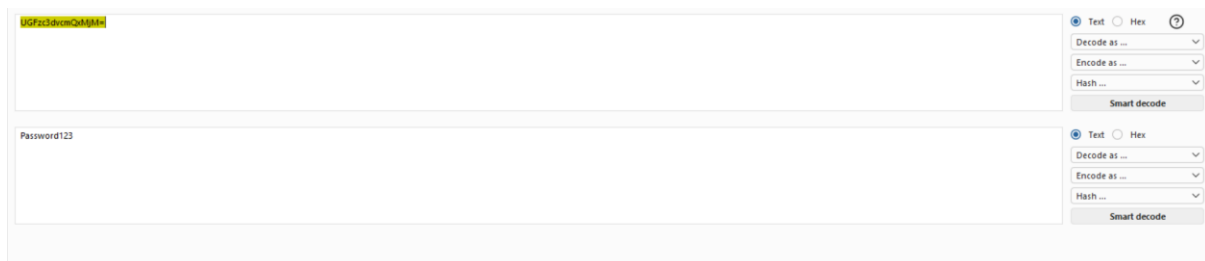
Output

Password123

Explanation

The Base64 encoded value represents “Password123”.

Screenshot



Conclusion

In this practical, Burp Suite Decoder and Comparer modules were successfully explored. Various encoding and decoding operations such as URL, Base64, and HTML encoding were performed. HTTP requests and responses were compared using Comparer, helping to identify similarities and differences in web traffic. This practical improved understanding of how web applications transmit and process encoded information.